Bi-weekly Report

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Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

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Objectives for this report

1. Innovative component by the team.

1 Basis

Report 5 closed the heat integration and the techno-economic analysis. The items open at that time which bear on this report are listed in Table 1.

Table 1: Position at the close of Report 5.

Item	Report 5 status	Taken up in
Purchased heating, 70 bar steam for VAP-1	1.80 MW, 4,307 kg/h	Sections 2, 4, 6
Reactor feed vapour fraction	0.988 at 160.2°C	Section 3
Bed peak with 40 bar steam on the reactor shell	Not checked	Section 5
Route to replace the 70 bar steam (thermal-oil loop proposed)	Proposed, not studied	Sections 2, 4
Letdown valves to 16 and 10 bar	Pressure energy not recovered	Section 8
Aniline purity, 99.9 wt% required	99.45 wt%	Section 11
Nitrobenzene in product, below 5 ppm	313 ppm	Section 11
Aniline and cyclohexylamine in wastewater	Carbon bed and biological treatment	Section 11

The flowsheet, property method (NRTL-RK with the regressed pairs R-DR-1 and DR-1) and heat exchanger network are those of Report 5. Only the feed heating section and the reactor steam drum pressure are changed.

Innovative component: stop boiling nitrobenzene on its own. Let the hydrogen loop carry it instead, by evaporating the nitrobenzene into the recycle and make-up hydrogen in a packed

saturator column heated by reactor steam. Then use the lower heating level this allows to reset the reactor steam drum pressure, so that the catalyst bed stays below its 400°C limit. Simulate the change, design the saturator, rebuild the steam balance, check the bed hot spot and evaluate the saving. Back-pressure turbines on the steam letdown were examined as a second option and are reported in Section 8.

1.1 Key results

In Report 5 the nitrobenzene is boiled alone at 263°C (3 bar). That single service needs 1.80 MW of purchased 70 bar steam, 4,307 kg/h or USD 1.24 million per year. Mixed with the hydrogen, the nitrobenzene has a partial pressure of only 0.29 bar, and Aspen Plus puts the dew point of the reactor feed at 163.9°C. The whole feed heating of 4.29 MW is then needed below 176°C, so reactor steam does all of it and VAP-1 and the 70 bar steam are removed.

The nitrobenzene is evaporated in a packed saturator rather than in a heat exchanger. Hot liquid nitrobenzene (176°C) runs down the packing against the rising hydrogen and evaporates into it; the gas leaves dry at 168.9°C, 3.8 K above its dew point. All the steam heat, about 4.2 MW, enters through a liquid-phase heater of 81 m^2, where a mixed gas–liquid heater would need 196 to 392 m^2. Nitrobenzene that the gas cannot carry stays in the column sump and cannot reach the catalyst.

Three defects of Report 5 are closed. The 70 bar steam import goes to zero. The reactor feed, which was 1.2 mol% liquid at 160.2°C (about 17 kmol/h, mostly nitrobenzene), becomes fully vapour. And the 40 bar steam drum of Report 5 (250.4°C on the shell) gives a bed peak of 422°C in a calibrated non-isothermal bed model, 22 K above the copper sintering limit. With nothing on the plant now needing heat above 176°C, the drum is reset to 33 bar (239.2°C) and the peak falls to 388°C with a 170°C feed, a margin of 12 K.

The net saving is USD 0.79 million per year for USD 0.49 million of added fixed capital (new saturator system less VAP-1 and the old E-102A). The incremental NPV at 12 per cent is USD 3.2 million, the IRR 83 per cent and the payback 0.8 years. The NPV stays positive in every single-variable sensitivity case, including export steam valued at full price. The back-pressure turbines give an NPV of about zero at the new 33 bar header and are not adopted.

1.2 Values carried over from Reports 4 and 5

The values carried over are in Table 2.

Table 2: Values carried over from Reports 4 and 5.

Item	Value (Reports 4 and 5)
Feed vaporiser VAP-1	Heater, 3 bar, vapour fraction 1, pure nitrobenzene; boils at 263°C
Cold stream S1 to S3 (E-102A, VAP-1, FEHE)	28.0 to 290.0°C, 3.89 MW
Feed heaters in the final network	E-102A 1.88 MW (40 bar steam, 54.8 m^2); VAP-1 1.80 MW (70 bar steam, 87.6 m^2); FEHE 0.21 MW (129.8 m^2)
Purchased heating	4,307 kg/h of 70 bar steam at USD 0.035/kg, USD 1.24 million per year
40 bar steam balance	Raised in E-101 15.93 MW; to E-102A 1.88, COL-1 4.06, COL-2 6.56; exported 3.43 MW (5,235 kg/h at USD 0.016/kg, half of full value)
Reactor feed after MIX-2	160.2°C, 3 bar, 1,452 kmol/h, nitrobenzene 141.89 kmol/h
Reactor (Report 4)	9700 tubes, 25 mm, 2.0 m; bed peak 390.7°C at a 138°C inlet; limit 400°C
Reactor shell (Report 5)	Saturated 40 bar steam, 250.4°C
Operating time	8200 h/y

The feed section of Report 5 has three problems, and this report is built around them.

1. **Purchased steam.** Nitrobenzene is boiled alone because it meets the hydrogen only later, at MIX-2. Pure nitrobenzene boils at 263°C at 3 bar, which is above the 40 bar reactor steam (250.4°C), so the plant buys 70 bar steam for this one service. It is the only purchased heating left after the heat integration of Report 5.
2. **Liquid to the catalyst.** The vaporised nitrobenzene at 290°C is mixed with cold hydrogen at MIX-2 and part of it condenses again. Aspen gives a vapour fraction of 0.988 for the reactor feed at 160.2°C, so about 1.2 mol% of the 1,452 kmol/h, roughly 17 kmol/h and mostly nitrobenzene, enters the bed as liquid. RXFEED in Reports 2 and 3 had vapour fractions of 0.990 and 0.987, so the defect has been present since Report 2.
3. **Bed temperature.** Report 4 checked the hot spot at 390.7°C for a coolant that it did not state. Report 5 then put saturated 40 bar steam (250.4°C) on the shell. Section 5 shows that this coolant temperature drives the peak to about 422°C, above the 400°C sintering limit.

2 The Idea: Let the Hydrogen Carry the Nitrobenzene

The reactor feed carries about 9 mol of hydrogen-rich gas per mol of nitrobenzene, so nitrobenzene only has to reach its dew point at its own partial pressure, not its boiling point at the full pressure. Its mole fraction in RXFEED is 141.89/1,452 = 0.0977, which at 3 bar is a partial pressure of 0.293 bar. A Clausius–Clapeyron line through the normal boiling point (210.8°C) and the Aspen boiling point at 3 bar (263°C) gives about 162°C at 0.293 bar, against 163.9°C from Aspen (Section 3). The heating level falls by about 100 K.

The question is how to evaporate it. Two arrangements were compared.

- **Mix, then heat (first draft of this study).** Liquid nitrobenzene is sprayed into the cold hydrogen and the two-phase mixture is heated in an enlarged E-102A. The heater is then a low-pressure gas heater whose coefficient is set by the gas film, so its area is uncertain (196 to 392 m^2 for U of 150 to 75 W/m^2K). Its outlet is still two-phase, because reaching the dew

point needs 4.18 MW and E-102A supplies 4.08 MW, so the FEHE takes a wet inlet. If the hydrogen flow drops, liquid nitrobenzene goes on towards the catalyst.

- **Saturator (adopted).** The heat is put into liquid nitrobenzene, which is pumped round a packed column and meets the hydrogen by direct contact. This is how process-gas saturators work in reforming and methanol plants, where hot water is evaporated into the feed gas.

The saturator was chosen for four reasons.

1. All the steam heat enters a liquid, at a coefficient of 500 to 800 W/m^{2}K instead of 75 to 150 W/m^{2}K, so the heater is 61 to 97 m^2.
2. The column is the phase separator. The gas leaves the top through a demister above its dew point, and any nitrobenzene the gas cannot carry stays in the sump, where a rising level shows it. Liquid cannot reach the bed unnoticed.
3. The hottest nitrobenzene on the plant falls from 290°C (VAP-1 outlet, then FEHE) to 176°C, and none of it boils on a steam-heated wall.
4. Heavy impurities in the feed nitrobenzene (dinitrobenzene, nitrophenols) collect in the sump and leave in a small blowdown instead of passing to the catalyst.

Once the feed needs nothing above 176°C, the highest heating level on the plant is the COL-1 reboiler at 185.95°C, served by 16 bar steam (201°C) let down from the reactor header. The reactor drum no longer has to make 40 bar steam, so it can be run at the pressure the catalyst needs. That second step is Section 5.

3 Aspen Plus Simulation

Aspen puts the reactor feed fully vapour at 163.94°C and 3 bar, needing 4.29 MW from the mixed feed to 170°C; the full recycle flowsheet with a 170°C feed leaves product and reactor results unchanged within convergence tolerance. Both tests were made in copies of the final Report 5 file, so NRTL-RK and its regressed binary pairs are unchanged.

3.1 Feed heating test (side branch)

The three feed streams were entered as copies of the converged streams S1, H2-MAKE and H2RET, with temperature, pressure and all component flows, so the branch is not connected to the recycles.

Table 3: Blocks of the test branch.

Block	Model	Specification
NEWNB, NEWH2M, NEWH2R	Streams	Copies of S1, H2-MAKE and H2RET from Report 5
MIXX	Mixer	3 bar; mixes the three streams
DEW	Heater	3 bar, vapour fraction 1: dew point and the duty to reach it
HEATY	Heater	3 bar, outlet 160.17°C (check) and 170°C (final)

Table 4: Aspen results for the feed with nitrobenzene evaporated into the hydrogen, 3 bar.

Case	Outlet T, °C	Vapour fraction	Duty from MIXX outlet, MW
DEW, vapour fraction 1 at 3 bar	163.94	1.000	4.183
HEATY at the Report 5 reactor-feed temperature	160.17	0.988	3.889
HEATY, final choice	170.0	1.000	4.290

Check of the copied streams. At 160.17°C the duty is 3.889 MW, equal to the 3.89 MW of S1 to S3 in Report 5. The same mixture at the same final state has the same enthalpy, so the copied streams are correct.

Why 170°C. The feed must reach the reactor above 163.94°C. A 170°C set point gives 6 K of superheat, and it is the temperature at which the bed was checked in Section 5. The duty rises by 0.40 MW over Report 5 (3.89 to 4.29 MW) because the reactor feed is 9.8 K hotter.

Dew point sensitivity. The margin moves with the hydrogen to nitrobenzene ratio and with pressure. Table 5 is a Clausius–Clapeyron line through the Aspen dew point and the normal boiling point of nitrobenzene; the same line gives 260.9°C for pure nitrobenzene at 3 bar against 263°C in Aspen.

Table 5: Dew point of the reactor feed.

Dew point, °C	2.9 bar	3.0 bar	3.1 bar	3.2 bar
H_2 10 % below design	165.8	167.0	168.1	169.2
Design (Report 4 feed)	162.8	163.9	165.1	166.2
H_2 10 % above design	160.1	161.2	162.3	163.4

The saturator top runs at 3.1 bar, where the design dew point is 165.1°C. At the reactor inlet (3.0 bar) the worst case in Table 5 is 167.0°C, so the 170°C set point keeps 3 K in hand when recycle hydrogen is 10 per cent low. A low alarm on the H_2 to nitrobenzene ratio is fitted for that case.

3.2 Full flowsheet test

A Heater was added between MIX-2 and REACT at 3 bar and 170°C, with VAP-1 and FEHE left as in Report 5, so the reactor, separation section and all four recycles respond to the new feed temperature. REACT here is the isothermal RPlug of Report 4 at 300°C.

Table 6: Full flowsheet with a 170°C reactor feed.

Item	Report 5	170°C feed	Change
Reactor feed temperature, °C	160.2	170.0	+9.8
Reactor feed vapour fraction	0.988	1.000	+0.012
Nitrobenzene into reactor, kmol/h	141.888	141.874	−0.014
Conversion per pass, %	99.49	99.50	+0.01
Selectivity to aniline, %	94.06	94.03	−0.03
Net duty removed by the coolant, MW	15.93	16.33	+0.40
Product flow, kg/h	12,185.8	12,182.9	−2.9
Aniline purity, wt%	99.45	99.44	−0.01
Nitrobenzene in product, ppm	313	306	−7

In an isothermal RPlug the feed is in effect brought to 300°C at the inlet, so its temperature does not enter the kinetics. The small changes in conversion, selectivity and product are the recycles re-converging (tear tolerance 0.001), not a reactor response. Two results from this run are real: the cooling duty rises by the 0.40 MW that the feed brings in, and the product is unchanged. The 24 t/y of product lost is still charged in Section 7, which is conservative. The effect of the hotter inlet on the bed is taken up with a non-isothermal model in Section 5.

4 Design of the Feed Saturator

A packed column 2.8 m across with 3.0 m of 50 mm Pall rings, a 160 t/h nitrobenzene pump-round and an 81 m^2 steam heater evaporates the whole feed and delivers the gas dry at 168.9°C. The design was calculated by hand (Python), not in Aspen; the Aspen check that remains is listed in Section 11.

4.1 Arrangement

- **C-101, saturator.** Hydrogen make-up and recycle gas (1,310 kmol/h, mixed at 54.7°C) enter below the packing. Hot nitrobenzene enters at the top through a liquid distributor. The gas leaves through a mesh demister at the top.
- **P-101 A/B, pump-round.** Liquid from the sump (135.8°C) is pumped through the heater back to the top. Fresh nitrobenzene (141.28 kmol/h at 28°C) and the COL-2 bottoms (NBRECYC) join the pump-round upstream of the heater, so the column receives a single liquid feed.
- **H-101, pump-round heater.** Shell and tube, condensing reactor steam on the shell, nitrobenzene in the tubes, 125.7 to 176°C. It replaces VAP-1 and the old E-102A.
- **FEHE (kept).** Superheats the saturator gas from about 169°C to the 170°C set point against reactor effluent, and covers the case where the gas leaves closer to its dew point (Section 4.3). Its duty falls from 0.21 MW to about 0.09 MW.
- **Blowdown.** A small continuous draw from the sump removes heavy residues; it is not quantified here because the feed specification of Report 1 gives no heavy-ends content.

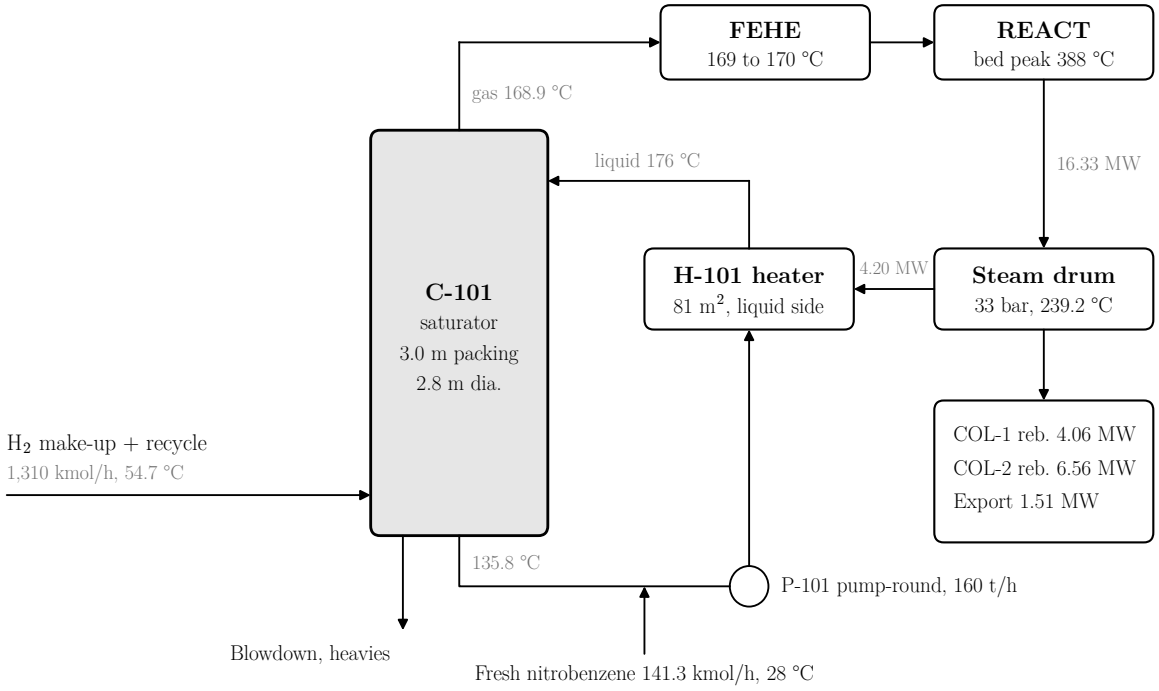


Figure 1: Proposed feed section and reactor steam header. The pump-round carries all the steam heat into the column as hot liquid; the gas is heated and saturated by contact, and the reactor drum at 33 bar supplies every heating duty on the plant.

4.2 Method

The column was treated as counter-current gas–liquid contact with the gas-side resistance controlling, since the liquid is nearly pure nitrobenzene and has no mass-transfer resistance of its own. The gas-phase balance was integrated up the packing in transfer units, with heat and mass transfer linked by the Chilton–Colburn analogy, and the liquid temperature taken at each height from the overall enthalpy balance. The bottom liquid temperature was found by shooting, so that the liquid reaches the heater outlet temperature exactly where all 141.89 kmol/h of nitrobenzene has entered the gas.

- Vapour pressure: the Clausius–Clapeyron line of Table 5 (latent heat 46.6 kJ/mol over this range).
- Gas: the Report 4 RXFEED components less nitrobenzene; ideal-gas heat capacities.
- Liquid nitrobenzene: $C_p = 186 + 0.17(T - 298)$ J/mol K.
- Lewis number of the gas: about 2.3, from a thermal diffusivity of about 5.5×10^{-5} m^2/s for the hydrogen-rich gas and a nitrobenzene–hydrogen diffusivity of about 2.4×10^{-5} m^2/s at 170°C and 3 bar. A value of 2.0 is used, so the heat transfer units are $\mathrm{Le}^{2/3} = 1.59$ times the mass transfer units.

Check against Aspen: the hand enthalpy balance gives 4.20 MW from the mixed feed to vapour at 170°C, against 4.29 MW in Table 4, that is 2 per cent low. The steam balance in Section 6 uses the Aspen total.

4.3 Results

Table 7: Saturator performance; selected case in bold. At 600 kmol/h and below the pump-round cannot carry the heat and the column pinches at the top.

Pump-round, kmol/h	Liquid to top, °C	Sump, °C	Gas out, °C	NTU (gas)	Heater duty, MW
1,000	172	119.1	167.5	5.36	4.16
1,000	176	123.5	168.7	3.64	4.18
1,300	172	131.5	167.8	3.79	4.16
1,300	**176**	**135.8**	**168.9**	**2.74**	**4.18**
1,300	180	140.3	169.4	2.12	4.19
2,000	176	150.1	168.6	2.07	4.18

The selected case circulates 1,300 kmol/h (160 t/h, about 150 m^3/h) and heats it to 176°C. The gas leaves at 168.9°C against a dew point of 165.1°C at 3.1 bar, so it is 3.8 K superheated and dry. The liquid is held 10 to 15 K above the gas at the top, which is the driving force for evaporation; a hotter top shortens the packing but raises the hottest nitrobenzene temperature, so 176°C was kept.

Sensitivity to the Lewis number. With $\mathrm{Le}^{2/3} = 1$, that is no credit for the high thermal diffusivity of hydrogen, the gas leaves at 162.9°C, 2.2 K below its dew point, as a fine mist. This is the reason the demister and the FEHE superheat are kept: the reactor sees 170°C dry gas whichever value is right. The Aspen check of Section 11 should settle it.

4.4 Sizing

Table 8: Saturator sizing.

Item	Basis	Result
Packing height	NTU 2.74 × HTU 0.6 m (50 mm metal Pall, gas film) × 1.8 for distribution	3.0 m
Column diameter	Eckert GPDC, $F_{LV} = 0.27$, $F_p = 89$ m^{-1}, 70 % of flood; gas 6.48 kg/s at 1.35 kg/m^3; liquid 49.3 kg/s at 1050 kg/m^3	2.8 m
Column height, tangent to tangent	Packing 3.0 m, sump 1.3 m, gas inlet 1.0 m, distributor and demister 1.5 m	7.0 m
Sump hold-up	3 minutes of pump-round	7.5 m^3
Gas pressure drop	Packing, distributor and demister	about 0.03 bar
H-101 heater	4.18 MW; 33 bar steam at 239.2°C against 125.7 to 176°C, LMTD 85.9 K; $U = 600$ W/m^2K (500 to 800)	81 m^2 (61 to 97)
H-101 steam	Latent heat 1769 kJ/kg at 33 bar	8,510 kg/h
P-101 A/B	150 m^3/h, 25 m head, η 0.75, motor 0.95	16 kW each, one spare

The hot nitrobenzene inventory is about 10 m^3 (sump, packing hold-up, heater and piping) at 126 to 176°C. It is a hazard to be reviewed, but a smaller one than the VAP-1 kettle of Report 5, which boils nitrobenzene at 263°C against 285.8°C steam.

4.5 Saturator against the mixed-feed heater

Table 9: Comparison of the two ways of evaporating into the hydrogen.

Item	Mix, then heat (E-102A enlarged)	Saturator (adopted)
Where the steam heat goes	Gas with liquid droplets, gas-film controlled	Liquid nitrobenzene
Overall coefficient, W/m^2K	75 to 150, uncertain	500 to 800
Steam-heated area, m^2	196 to 392	61 to 97
State leaving the steam heater	Two-phase, about 162.6°C	Liquid; gas leaves C-101 dry at 168.9°C
If H_2 falls 10 per cent	Liquid nitrobenzene carried on to the FEHE and reactor	Excess stays in the sump; level alarm
Hottest nitrobenzene	Film on a 250°C tube wall	176°C liquid, never boiled on a wall
Heavy residues	Carried to the catalyst	Removed by sump blowdown
Extra gas pressure drop	About 0.15 bar	About 0.08 bar including FEHE
Added equipment cost, CBM, USD million	0.25 to 0.41 for E-102A, plus a knock-out drum	0.62 for the whole system (Table 13)

The saturator costs more to build, by roughly USD 0.2 to 0.3 million of bare-module cost. It is preferred because the result no longer depends on an uncertain gas-side coefficient, and because no upset can send liquid nitrobenzene to the bed.

5 Reactor Hot Spot and the Steam Drum Pressure

The bed peak is set by the coolant, not by the feed: 10 K more at the inlet adds 1 K at the peak, while 1 K more on the shell adds 3.2 K. On the Report 5 drum (40 bar, 250.4°C) the peak is 422°C, so the drum is reset to 33 bar (239.2°C), which gives 388°C with the 170°C feed.

5.1 Non-isothermal bed model

The flowsheet reactor is an isothermal RPlug, so it cannot show the axial profile. One tube of the Report 4 bed was therefore modelled as a one-dimensional plug flow with the Report 4 kinetics (Table 2 of Report 4), geometry (25 mm, 2.0 m, 9700 tubes, voidage 0.4), the Report 4 pressure drop of 0.279 bar, and heat removal $U\pi d\,(T - T_c)$ to the shell.

- **Rate basis.** Run isothermal at 300°C with rates on the fluid (void) volume and 0.131 bar, the model gives 99.49 per cent conversion and selectivities of 94.07, 1.75 and 4.18 per cent. The converged Aspen RPlug gives 99.49, 94.06, 1.76 and 4.19. The kinetics and rate basis are therefore those Aspen uses.
- **Heat of reaction.** −473 kJ/mol for the main reaction, so that the bed releases the 18.75 MW of Report 4; −193 and −51 kJ/mol for the side reactions (Report 1).
- **Calibration.** Report 4 does not state U or the coolant temperature. They were fitted to the Report 4 Aspen case (peak 390.7°C, outlet 242.5°C at a 138°C inlet), giving $U = 505$ W/m^2K and $T_c = 240.8$°C.
- **Checks.** The peak sits at 0.21 m against 0.293 m in Aspen. For the 52.5 mm tubes the model

gives 670°C against 640.4°C in Report 4, 4.6 per cent high.

The absolute values are therefore indicative; the result used is how the peak moves with the inlet and coolant temperatures.

5.2 Results

Table 10: Bed model results, 25 mm tubes. Selectivities from the bed model, not the flowsheet.

Case	Inlet, °C	Coolant, °C (drum)	Peak, °C	Margin to 400°C, K	Outlet, °C	Conversion, %	S aniline, %	S cyclohexylamine, %
Report 4 calibration	138.0	240.8	390.7	+9.3	242.5	97.4	96.7	0.98
Report 5 as designed	160.2	250.4 (40 bar)	422.2	−22.2	251.6	98.4	95.2	1.28
170°C feed, 40 bar drum	170.0	250.4 (40 bar)	423.2	−23.2	251.6	98.4	95.2	1.29
170°C feed, 34 bar drum	170.0	240.9 (34 bar)	393.5	+6.5	242.5	97.5	96.6	1.00
This report, 33 bar drum	**170.0**	**239.2 (33 bar)**	**388.0**	**+12.0**	**240.9**	**97.3**	**96.8**	**0.95**

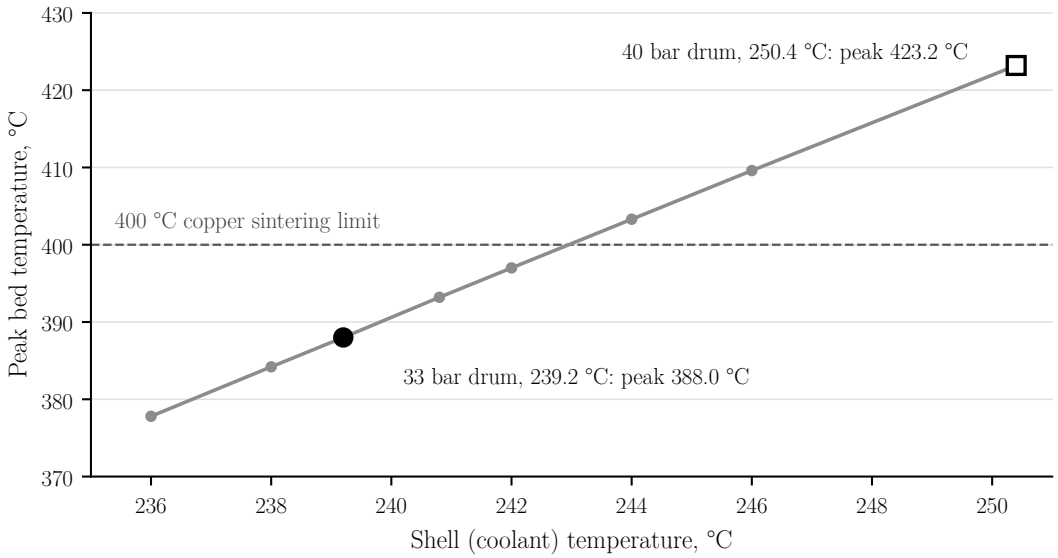


Figure 2: Bed model, 170°C feed, 25 mm tubes. The peak rises 3.2 K for every kelvin on the shell, so the drum pressure, not the feed temperature, decides whether the bed stays below 400°C.

The Report 5 drum is too hot. Report 4 checked the hot spot with a coolant near 241°C, and Report 5 then fixed the shell at 250.4°C to raise 40 bar steam. On the calibrated model that puts the peak 22 K over the limit. The defect is in Report 5 and is independent of the feed temperature.

Why 33 bar. At 34 bar the margin is 6.5 K. The peak moves 3.2 K for every kelvin of shell temperature, and a ±0.5 bar swing in drum pressure moves the shell by ±0.85 K, that is ±2.7 K at the peak. 33 bar keeps 12 K of margin, more than the 9.3 K that Report 4 accepted. Nothing on the plant needs the steam hotter: the hottest user is the COL-1 reboiler at 185.95°C on 16 bar letdown steam (201°C), and H-101 heats to 176°C.

Why the change is free only with the saturator. In Report 5, E-102A preheats nitrobenzene to 240.4°C on the 40 bar header. A 33 bar header (239.2°C) cannot do that, so in the old scheme the drum reset would move about 0.1 MW more onto purchased 70 bar steam. With the saturator no user is above 176°C and the reset costs nothing.

Effect on the reaction. The cooler shell also lowers the bed temperature after the peak, so the model gives less cyclohexylamine (0.95 against 1.28 per cent) and more aniline (96.8 against 95.2 per cent). Less cyclohexylamine helps the 99.9 wt% purity target, which cyclohexylamine alone holds back. Conversion falls from 98.4 to 97.3 per cent, and the extra 1.6 kmol/h of nitrobenzene returns through the COL-2 bottoms. Neither effect is credited in Section 7, because the 1-D model gives a lower conversion than the isothermal flowsheet in every case, including the Report 4 calibration case. This is a property of the Report 4 bed and needs a non-isothermal RPlug run inside the recycle flowsheet (Section 11).

6 Steam Balance and Heat Integration

The reactor header now carries every heating duty on the plant and still exports 1.51 MW, with no purchased steam. The reactor feed brings 0.40 MW more heat in, and the isothermal reactor removes that much more through E-101 (Table 6). H-101 takes 4.20 MW of the 4.29 MW feed heating and the FEHE the remaining 0.09 MW.

Table 11: Steam balance with and without the saturator. Export kg/h scaled from Report 5 on the same duty basis (3.43 MW equals 5,235 kg/h).

Item, MW	Report 5	This report	Change
Drum pressure, saturation temperature	40 bar, 250.4°C	33 bar, 239.2°C	−7 bar
Steam raised in E-101	15.93	16.33	+0.40
Feed heating from the header	1.88 (E-102A)	4.20 (H-101)	+2.32
COL-1 reboiler, via 16 bar letdown	4.06	4.06	0
COL-2 reboiler, via 10 bar letdown	6.56	6.56	0
Reactor steam exported	3.43	1.51	−1.92
Reactor steam exported, kg/h	5,235	2,305	−2,930
70 bar steam bought for VAP-1	1.80 (4,307 kg/h)	0	−1.80
FEHE	0.21	0.09	−0.12
10 bar steam raised in E-104	2.17	2.29	+0.12
10 bar steam exported, kg/h	1,932	2,116	+184

In effect, 1.80 MW of bought 70 bar steam is replaced by 1.80 MW of reactor steam that was being exported, and a further 0.12 MW of exported reactor steam becomes 10 bar steam from E-104, because the FEHE takes less from the reactor effluent. The specific enthalpy of saturated steam is almost the same at 33 and 40 bar (2803 and 2801 kJ/kg), so the export kg/h is unchanged by the drum pressure; only its temperature level is lower.

The header keeps 1.51 MW (about 2.3 t/h) in hand, 9 per cent of the steam raised. That margin is what limits combining this change with the turbines of Section 8.

Effect on the network of Report 5. VAP-1 and E-102A are removed and H-101 is added, so the exchanger count falls from 14 to 13 and C-101 is added. H-101 is heated by reactor steam that Report 5 exported, so no process-to-process match of the Report 5 network is disturbed. Purchased heating falls to zero, which is the minimum hot utility that Aspen Energy Analyzer gave as the target in Report 5.

7 Economic Evaluation

The change saves USD 0.79 million per year for USD 0.49 million of added fixed capital, an incremental NPV of USD 3.2 million at 12 per cent. Utility prices, the export credit, CEPCI 797.9 and the cash flow method are those of Report 5.

7.1 Annual saving

Table 12: Annual saving at the utility prices of Report 5.

Item	Basis	USD million/y
70 bar steam no longer bought	4,307 kg/h × 8200 h × USD 0.035/kg	+1.236
Reactor steam export lost	2,930 kg/h × 8200 h × USD 0.016/kg	−0.384
Extra 10 bar steam exported	184 kg/h × 8200 h × USD 0.013/kg	+0.020
Boiler feed water on extra steam raised	796 kg/h × 8200 h × USD 0.0005/kg	−0.003
Aniline revenue lost (Table 6)	24 t/y × USD 1,632/t	−0.039
Extra electricity	22 kW recycle compressor (0.08 bar more) + 16 kW pump-round = 38 kW × 8200 h × USD 0.077/kWh	−0.024
Maintenance on the added capital	3 % of USD 0.486 million	−0.015
Net annual saving		**+0.791**

The compressor term scales the 213 kW of COMP (Report 4) for 0.08 bar more discharge pressure, isentropic with $\gamma = 1.4$. Plant demand rises from about 570 to 608 kW, still within the 757 kW gas turbine. Electricity is valued at USD 0.077/kWh, the fuel cost of gas-turbine power (Report 5).

7.2 Capital cost

Equipment is costed with the bare-module correlations of Turton et al., CEPCI 397 (2001) to 797.9 (2023), and the grass-roots factor of Report 5, fixed capital = 1.68 C_{BM}.

Table 13: Capital cost of the change. Study estimate, about ±30 per cent.

Item	Size	C_{BM}, USD million
C-101 saturator shell	2.8 m × 7.0 m, carbon steel, 2.1 barg	0.307
C-101 packing	18.5 m^3 of 50 mm Pall rings, 304 stainless	0.070
H-101 pump-round heater	81 m^2, floating head, 33 bar shell	0.165
P-101 A/B	16 kW centrifugal, carbon steel, one spare	0.077
New equipment		**0.619**
Removed: VAP-1	87.6 m^2, 70 bar shell	−0.181
Removed: E-102A of Report 5	54.8 m^2, 40 bar shell	−0.148
Net change in bare-module cost		**0.290**
Net added fixed capital, 1.68 C_{BM}	1.039 new less 0.553 removed	**0.486**

The plant has not been built, so the exchangers removed from the Report 5 design are credited at the same factor as the equipment added. The case with no credit is in Table 14. The steam drum and reactor shell could also be rated for 33 bar instead of 40 bar; that small saving is not taken. No working capital is added, as the change holds no extra inventory of value.

7.3 Profitability and sensitivity

The cash flow follows Report 5: two construction years at 60 and 40 per cent, 15 years of operation, straight-line depreciation over 10 years, 25 per cent tax, 12 per cent discount rate, no salvage. As a check, the same routine reproduces the NPV of the first draft of this study (USD +3.52 million) and of the turbine draft (USD −0.14 million).

Table 14: Incremental profitability and sensitivity, 12 per cent, 15 years.

Case	Saving, USD M/y	Added capital, USD M	NPV, USD M	IRR, %	Payback, y
Base case	**0.791**	**0.486**	**+3.20**	**83**	**0.8**
No credit for VAP-1 and E-102A	0.774	1.039	+2.67	46	1.7
New equipment cost +50 %	0.775	1.006	+2.70	47	1.7
Export steam valued at full price (USD 0.032/kg)	0.406	0.486	+1.45	50	1.5
No buyer for the export steam	1.175	0.486	+4.96	111	0.5
70 bar steam price halved	0.173	0.486	+0.38	24	3.4
70 bar price halved and equipment +50 %	0.157	1.006	−0.12	10	7.0

The NPV is positive in every single change, including the adverse one where the lost export steam is worth its full price. It goes negative only when the 70 bar steam is half the Report 5 price and the new equipment costs 50 per cent more at the same time. The change adds about USD 3.2 million to the plant NPV of USD 61.3 million in Report 5, about 5 per cent. The bed-selectivity gain of Section 5 is not counted; if the 1.6 point rise in aniline selectivity were confirmed in Aspen it would be worth more than the whole steam saving.

8 Also Examined: Back-Pressure Turbines on the Letdown

Replacing the two letdown valves with back-pressure turbines gives an incremental NPV of about zero and is not adopted. In Report 5 the reactor steam is let down through valves to 16 bar for COL-1 and to 10 bar for COL-2, and its pressure energy is lost. Turbines in the same positions recover it as power while the reboilers still get their 4.06 and 6.56 MW.

8.1 Turbine study on the Report 5 header (40 bar)

The steam system was simulated in a separate Aspen file with water only and STEAMNBS (turbine case Che453_Inn.apw, valve case Che453_Inn_valve.apw). T101 and T102 are Compr blocks in turbine mode, isentropic efficiency 0.75 and mechanical efficiency 0.95. Heaters with zero pressure drop and an exit vapour fraction of 0 stand for the reboilers, and two design specifications vary the steam flows until the duties are met.

Table 15: Turbine results from Aspen Plus at 40 bar.

Quantity	T101 (40 to 16 bar)	T102 (40 to 10 bar)	Total
Steam flow, kg/h	8,081	12,824	20,905
Exhaust vapour fraction	0.935	0.915	–
Indicated power, kW	297.3	693.0	990.2
Brake power, kW (× 0.95)	282.4	658.3	940.7
Electrical power, kW (× 0.96)	271.1	632.0	903.1

The valves need 19,130 kg/h of 40 bar steam; the turbines need 20,905 kg/h, 1,776 kg/h (9.3 per cent) more, because the turbine exhaust is wet. IAPWS-IF97 gives isentropic enthalpy drops from saturated 40 bar steam of 174.3 kJ/kg to 16 bar and 257.2 kJ/kg to 10 bar. At 0.75 efficiency these give 293 and 686 kW indicated, within 1.5 per cent of Aspen, and exhaust qualities of 0.937 and 0.916 against 0.935 and 0.915. The 174.3 kJ/kg figure was rechecked because an earlier review suggested 185 kJ/kg; the steam tables support 174.3.

The economics carry the corrections made after review: the generator efficiency of 0.96 used for the gas turbine in Report 3; USD 0.25 million for the generator, switchgear, synchronising panel and grid connection; the gas-turbine credit at the same 1.68 factor as the new turbines; USD 0.08 million to lengthen the thermal oxidiser chamber from 4.8 to 5.6 m so that it can take the 0.61 MW purge and keep 1 s at 870°C; and a grid standby charge of USD 0.036 million per year.

Table 16: Turbine study, incremental economics at 40 bar (12 per cent, 15 years).

Case (40 bar header)	Benefit, USD M/y	Capital, USD M	NPV, USD M	IRR, %	Payback, y
Base, gas turbine retired, export USD 0.05/kWh	0.349	2.086	−0.14	10.7	6.7
Export price USD 0.08/kWh	0.431	2.086	+0.23	13.9	5.6
No buyer for the 40 bar steam	0.582	2.086	+0.92	19.3	4.3
Isentropic efficiency 0.60	0.320	2.086	−0.27	9.5	7.1
Gas turbine kept running, all turbine power exported	0.137	3.912	−2.63	–	–

8.2 Turbines with the saturator and the 33 bar drum

The two options compete for the same steam, so the turbines were recalculated on the header of this report with IAPWS-IF97 at the same efficiencies.

Table 17: Turbines on the two headers. The 33 bar column is a steam-table calculation, not an Aspen run.

Item	40 bar header	33 bar header
Isentropic drop to 16 / 10 bar, kJ/kg	174.3 / 257.2	139.3 / 223.9
Electrical power, kW	892 (903 in Aspen)	747
Extra header steam over the valves, kg/h	1,754 (1,776 in Aspen)	1,465
Header steam left for export, kg/h	–	2,305 − 1,465 = 840
Benefit, USD M/y	0.349	0.310
Net added capital, USD M	2.086	1.728
Incremental NPV, USD M	−0.14	−0.02

At 33 bar the turbines still pay for themselves only at break-even. They would also leave 840 kg/h, about 3 per cent of the steam raised, as the only margin for turndown or a reboiler upset, and their power would follow the reboiler load, so a column trip would take out the recycle compressor and the cooling-water pumps unless the grid picks them up. The turbines become worth installing only where the export steam has no buyer (NPV about +0.5 million at 33 bar) or electricity sells above about USD 0.06/kWh (the 40 bar break-even). They are recorded as an option for that case and are not part of the design.

9 Discussion

- **Technical.** The feed is fully vapour above 163.9°C at 3 bar, and the saturator delivers it at 168.9°C before the FEHE trims it to 170°C. The separation section and the recycles see the same product (Table 6) and need no change. The saturator, its pump-round and H-101 replace two exchangers, so the plant has one more vessel and one fewer exchanger than in Report 5.
- **Safety, two effects in opposite directions.** Nitrobenzene is no longer boiled at 263°C and superheated to 290°C, and the hottest liquid nitrobenzene is 176°C, which reduces the thermal decomposition hazard on the feed side. The reactor inlet, on the other hand, is 9.8 K hotter, which raises the hot spot and the runaway risk in the bed by about 1 K. The drum reset to 33 bar more than covers that: the peak falls from 422°C (Report 5 as designed) to 388°C. The bed check is therefore a safety item, not only a performance item. The hazard review must also cover the 10 m^3 of hot nitrobenzene in the saturator loop and nitrobenzene vapour in a hydrogen-rich gas.
- **Control.** Four loops matter: the H-101 steam valve holds the liquid to the top at 176°C; the FEHE bypass holds the reactor feed at 170°C; the drum pressure controller holds 33 bar and so the shell at 239.2°C; and the sump level shows the nitrobenzene balance. A rising level means the gas is not carrying all the feed (low hydrogen, low liquid temperature), and it is alarmed together with the H_2 to nitrobenzene ratio.
- **Start-up.** The pump-round is heated on steam before the hydrogen flow is established, and nitrobenzene feed is admitted only when the gas leaving C-101 is above its dew point. Reactor steam is not available before the reaction lights off, so H-101, like the column reboilers of Report 5, needs a start-up steam supply.

- **Combined with the turbines.** The turbines would take 1,465 of the 2,305 kg/h of steam the header still exports and leave about 3 per cent of the steam raised as margin. They are not adopted (Section 8); if they were, a standby steam source would be needed.

10 Self-Assessment

1. The saving rests on the 70 bar steam price of Report 5. Halving it still leaves a positive NPV of USD 0.38 million.
2. The drum reset is the more important safety result of this report. It rests on a calibrated model whose absolute values carry a few per cent of error; the 12 K margin is chosen so that the decision holds within that error.
3. The saturator design depends on an estimated Lewis number. The FEHE superheat and the demister are kept so that the reactor feed is dry either way.
4. The turbine option was studied fully and is reported as not worth installing at present prices.

11 Objectives Which Could Not Be Accomplished, With Reasons

The objective of this report, the innovative component, is complete. The parts of the study still open are given first. The product and discharge specifications set in Report 1 are separate from this objective and are carried forward; their position is given after them.

The saturator is not yet in Aspen. It was designed by hand. The check is a RadFrac with no condenser or reboiler, 4 to 6 equilibrium stages, gas to the bottom stage, a pump-round from the bottom to the top through a Heater at 176°C, and fresh nitrobenzene into the pump-round; it would replace the 2 per cent gap in duty and settle the outlet superheat of Section 4.3.

The bed check is a calibrated 1-D model, not the Aspen RPlug. The model reproduces the isothermal Aspen reactor exactly and the Report 4 hot spot by calibration, but its conversion is lower than the flowsheet's. A non-isothermal RPlug with a 239.2°C coolant, inside the recycle flowsheet, is needed to confirm the peak and to decide whether the selectivity gain of Section 5 is real.

The mixing and saturator section is not drawn in the full flowsheet. Only the reactor feed temperature was changed there (Section 3.2).

Heavy ends and blowdown. The Report 1 feed specification gives no heavy-ends content, so the sump blowdown is not sized.

Costs are study estimates (±30 per cent) and the U of H-101 is a range, not a rating calculation.

Aniline purity. 99.44 wt% with the 170°C feed (Table 6), against 99.45 wt% in Report 5 and a target of 99.9 wt%. Cyclohexylamine holds it back, as in Report 5; the bed model of Section 5 indicates that the 33 bar drum lowers its formation.

Nitrobenzene in the product. 306 ppm with the 170°C feed, against 313 ppm in Report 5 and a target of 5 ppm. A taller COL-2 remains the route, as in Report 5.

Aniline and cyclohexylamine in the wastewater. Unchanged by this report; they are handled by the carbon bed and the biological treatment of Report 5.

12 Any Other Challenges

The first arrangement was replaced. The first draft of this study mixed the liquid nitrobenzene into the cold hydrogen and heated the two-phase stream. The heater area then depended on an uncertain gas-film coefficient and its outlet stayed two-phase, so the saturator was developed in its place (Section 4.5).

The saturator pinched at a low pump-round. At 600 kmol/h and below the pump-round could not carry the heat and the column pinched at the top; the pump-round was raised to 1,300 kmol/h (Table 7).

The rate basis of the kinetics. With the rates on the whole tube volume the isothermal bed model gave almost complete conversion and a selectivity to aniline near 85 per cent, far from Aspen. On the fluid (void) volume it reproduces the Aspen reactor (Section 5.1).

The isothermal reactor cannot show the inlet effect. The flowsheet RPlug is held at 300°C, so the effect of the 170°C feed on the hot spot had to be found with the separate bed model of Section 5.

The turbine hand check. An earlier review gave 185 kJ/kg for the isentropic drop from 40 to 16 bar. The steam tables give 174.3 kJ/kg, which was kept (Section 8.1).

13 Number of Hours Spent During This Period

Approximately 8 hours per member, that is about 48 person-hours for the group.

14 Contributions From Individual Members

Nithin T M and Aditya Nitin Patil — the Aspen side branch (MIXX, DEW, HEATY), the full flowsheet run at 170°C, the dew point study, and the compilation of this report.

Nithin D H and Shubham Singh — the turbine and valve cases in Aspen with the steam-table checks, the steam balance, the capital cost, and the cash flow and sensitivity cases.

Shivam Dongare and Sajag Masane — the saturator design and sizing, the non-isothermal bed model and drum pressure study, and the review of the two first drafts.

All six members have reviewed the complete report.

15 References

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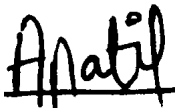
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Signatures of members



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